Introduction to Unix and R

Prerequisite: None

Recurring Zoom Link: See Shared Course Calendar

Instructor:

Paul Lott, PhD (Genetics)
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Course Schedule: Every Monday & Wednesday 5:00 to 6:30PM PDT (UTC−07:00) until June 3rd

Course Length: 10 Weeks

Course Chat/Help Forum:

Slack Channel: https://ucdavis.slack.com/archives/C0AMVCD6FS9

Slack Email: d43_unix_r-aaaatrlfzy2dkazdpi5zvc46we@ucdavis.slack.com

Course Overview:

This course is meant to be a low-stress introduction to Unix and R/Rstudio for laboratory researcher and medical professionals.

This course presents the basics of Unix Command-Line and R programming. The course is designed to get you ready to utilize large research clusters. We'll be using the UCDavis HIVE Cluster for our course – so students will be familiar with the do's and don'ts of working on a shared high performance cluster (HPC) environments.

In the first 3 weeks of this course, we'll cover the basics of Unix: filesystem, commands, pipes and streams, and process management. This will prepare you with the skills to login, navigate, utilize, and perform analysis using a HPC.

Then for Weeks 4 through 10, we'll switch our focus to using R and Rstudio on the HPC. R/Rstudio is a very commonly used analysis tool and programming language used in bioinformatics analysis. We'll cover the basics of R - datatypes, functions, syntax, and programming fundamentals. Students will become familiar with Unix, writing R code in Rstudio Markdown documents and troubleshooting code. We'll also cover some of the basic resources for R for conducting exploratory data analysis – including generating publication ready plots.

Course Requirements: Students are expected to have access to a laptop or desktop computer that also has a webcam for zoom classes. In order to access Zoom classes and computational resources, the student should plan to be in a location with stable internet access.

Course Objectives: Upon successful completion of this course students will be able to:

1. Access UCDavis HIVE HPC Infrastructure
2. Access and Use Unix Terminal
3. Navigate and manage a Unix file systems
4. Utilize Unix command-line applications to:
    a. View, filter, and parse files by utilizing Unix piping
5. Run Apps and understand parameters/arguments
6. Connect to a remote server via SSH
    a. Transfer files using SCP, SFTP, Rsync
7. Demonstrate a familiarity with Unix, R and RStudio
8. Generate a well-documented R Markdown Workbook
9. Identify fundamental computational datatypes
10. Use appropriate datatypes to store and analyze data
11. Create reusable functions that utilize flow control and loops
12. Utilize data science tools such as Tidyverse to:
    a. Import and export data
    b. Generate plots of data
    c. Perform basic statistical analysis of real world datasets.
13. Be able to debug error messages and troubleshoot basic coding problems
14. Solve programming problems by breaking them down into smaller steps

Participation: This course is an interactive course. Student will be expected to participate, ask clarifying questions, execute code in real-time, and help solve programming challenges/exercises. Exercises will be completed with the class in real-time.

Weekly assignments will be given and will be due the following Sunday night. The assignments may be completed individually or collaboratively. If help on assignments is required, participants should ask for help from others (students, colleagues, instructors) and utilize AI as a lost resort. We will review solutions to Assignments on the Monday following when we covered the section where an assignment was posted.

Zoom Etiquette: This course is meant to be an interactive class. To facilitate interaction, students should keep their cameras on, unless they need to step away briefly for personal reasons. Everyone will maintain a respectful environment for the other participants – this includes muting your microphone, when you are listening and only un-muting to ask questions, make comments, or interact. This helps to keep background noise to a minimum and helps others to be able to be attentive.

Because this is a very interactive class, we **won't be using "Raised Hands"**. Rather, students are encouraged to un-mute anytime they want to interact. If two or more students want to speak at once, please be considerate and take turns – so as not to speak over top of one another.

Schedule:

Date	Subject	Assignment
3/30/2026 (Week 1)	• Introductions **Unix:** ○ Introduction to Unix ○ Getting Started on Hive ○ Navigating Filesystem	None
4/1/2026 (Week 1)	**Unix:** ○ Managing the Filesystem	Assignment 1: Generate a directory tree
4/6/2026 (Week 2)	**Review Week 1 Assignment** **Unix:** ○ File permissions ○ Viewing File Contents ○ Stream Redirection and Pipes	
4/8/2026 (Week 2)	**Unix:** ○ Stream Redirection and Pipes cont. ○ Unix Power Tools	Assignment 2: Cleaning Penguins
4/13/2026 (Week 3)	**Review Week 2 Assignment** **Unix:** ○ Unix Power Tools Cont. ○ Process and Resource Management ○ SSH (Secure Shell)	Assignment 3: Cleaning PLINK File & Convert VCF to BED
4/15/2026 (Week 3)	**Unix:** ○ SSH (Secure Shell) ○ Customizing you Unix Profile	Assignment 3.1: SSH Login, SFTP File Transfer, Alias
4/20/2026 (Week 4)	**Review Week 3 Assignment** **R:** ○ Introduction to R & RStudio ○ Installing Libraries ○ R Markdown	
4/22/2026 (Week 4)	**R:** ○ R Markdown Cont. ○ Accessing Files, Directories, Basic	Assignment 4: Create R Project and RMarkdown
4/27/2026 (Week 5)	**Week 4 Assignment Review** **R:** ○ Datatypes & Strings ○ Vectors, Lists, Arrays ○ Basic Plotting	
4/29/2026 (Week 5)	**R:** ○ Matrices ○ Dataframes	Assignment 5: Plotting Penguins and Age
5/4/2026 (Week 6)	**Week 5 Assignment Review** **R:**	

	○ Programming with R ○ Conditionals	
5/6/2026 (Week 6)	**R:** ○ Loops ▪ For ▪ While	Assignment 6: Using loops to process datatypes
5/11/2025 (Week 7)	**Review Week 6 Assignment** **R:** ○ Functions ○ First Functions	Assignment 7: Calculate GC Percentage with R String functions
5/13/2025 (Week 7)	**R:** ○ Introduction to Tidyverse ▪ Readr ▪ Stringr ▪ Purr ▪ lubridate	Assignment 7.1: Translate DNA to Protein Assignment 7.2: Pattern Matching
5/18/2025 (Week 8) *Ascension Day??*	**Review Week 7 Assignment** **R:** ○ TidyVerse: Import	
5/20/2025 (Week 8)	**R:** ○ TidyVerse: Tidy	Assignment 8 : Import and Tidy Dataset
5/25/2025 (Week 9)	**Review Week 8 Assignment** **R:** ○ TidyVerse: ▪ Transform ▪ Visualize ▪ Model	
5/27/2025 (Week 9)	**R:** ○ TidyVerse: ▪ Transform ▪ Visualize ▪ Model	Assignment 9 : Transform, Visualize, Model Dataset
6/1/2025 (Week 10)	**Review Week 9 Assignment** **R:** ○ Statistical Analysis	None
6/3/2025 (Week 10)	**R:** ○ Statistical Analysis	None

Keys for Success: Students can ensure success in this course by doing the following:

1. Attend all classes
2. Follow along and perform the exercises with the instructor in real-time
3. Ask questions
4. Review and practice a little every day.
5. Work on assignments a little every day – don't wait until Sunday evening to try to complete the assignment.
6. Use the recommended and provided resources in the lessons.
7. Search out additional resources (Cheatsheets, Videos, Tutorials, Documentation, Help)
8. Ask for help

Guide to Getting Help: It can be a little frustrating when things don't work out as we hope or thought they should. Every programmer has dealt with this – even the experts. **Don't give up!!** Here are tried and tested methods to help effectively get help:

1. When you start programming test each and every step of your program: It's easier to identify where the problem is when you've already tested each previous line of code.
2. If you get stuck, do the following:
    a. **Try to figure it out yourself (10-15 Minutes):**
        i. Read the error message
        ii. Read your code
        iii. Double check your assumptions
            1. Print the variable
            2. Read documentation on the function
            3. Try the function on something that is known or an example.
        iv. Take a step away – often taking a step away for a few minutes can help you see the issue when you come back.
    b. **Get help from AI Coding Agent**
    c. **Google your error or issue (5-10 Minutes)**
        i. Often times someone else has experienced something similar or made a similar mistake – Google is a good place to search for help
    d. **Ask for help:**

i. Reach out for help
    1. If you're emailing, provide a copy of your code and explanation of what you're trying to do and what error you're running into.
    2. If you're zooming, be ready to share your screen and demonstrate the error and run code or print variables.

Zoom Session Recordings

Day and Lesson	**Zoom Recording Link**
Introductions and Unix Lesson 1	

Recommended Unix Tutorials:

https://users.cs.duke.edu/~alvy/courses/unixtut/

https://www.geeksforgeeks.org/linux-tutorial/

https://linuxjourney.com/

Recommended R Tutorials:

https://www.w3schools.com/r/

https://www.statmethods.net/r-tutorial/index.html

https://www.geeksforgeeks.org/r-tutorial/

https://education.rstudio.com/learn/beginner/

https://swirlstats.com/

https://intro2r.com/

https://rstudio.github.io/cheatsheets/contributed-cheatsheets.html

Additional Tutorials & Books:

- **R for Data Science:** https://r4ds.hadley.nz/ (Espanol)
- **Guide to R:** https://bookdown.org/yih_huynh/Guide-to-R-Book/tidyverse.html
- **Statistical Inference via Data Science:** https://moderndive.com/
- **R Graphics Cookbook:** https://r-graphics.org/recipe-bar-graph-adjust-width
- **Text Mining with R:** https://www.tidytextmining.com/
- **R Markdown Cookbook:** https://bookdown.org/yihui/rmarkdown-cookbook/
- **R Markdown - The Definitive Guide:** https://bookdown.org/yihui/rmarkdown/
- **R Programming for Data Science:** https://bookdown.org/rdpeng/rprogdatascience/
- **Efficient R Programming:** https://csgillespie.github.io/efficientR/
- **Advanced R:** https://adv-r.hadley.nz/
- **Fundamentals of Data Visualization:** https://clauswilke.com/dataviz/
- **Hands-on Programming with R:** https://rstudio-education.github.io/hopr/
- **bookdown - Authoring Books and Technical Documents with R Markdown:** https://bookdown.org/yihui/bookdown/
- **Big Book of R:** https://www.bigbookofr.com/
- **RStudio Resource Books:** https://www.rstudio.com/resources/books/
- **Bookdown.org:** https://bookdown.org/
- **Rstudio Learn:** https://education.rstudio.com/learn/beginner/

Libros tutoriales y hojas de trucos en español

- BaseR Cheatsheet
- Descriptive Statistics Cheatsheet
- R Markdown Cheatsheet
- RStudio Cheatsheet
- R Syntax Comparison
- CRAN Documentation
    - **"R para Principiantes"**, the Spanish version of "R for Beginners", translated by Jorge A. Ahumada (PDF).
    - A Spanish translation of **"An Introduction to R"** by Andrés González and Silvia González (PDF, Texinfo sources).
    - **"Gráficos Estadísticos con R"** by Juan Carlos Correa and Nelfi González (PDF).
    - **"Cartas sobre Estadística de la Revista Argentina de Bioingeniería"** by Marcelo R. Risk (PDF).
    - **"Introducción al uso y programación del sistema estadístico R"** by Ramón Díaz-Uriarte, transparencies prepared for a 16-hours course on R, addressed mainly to biologists and bioinformaticians (PDF).
    - **"Generacion automatica de reportes con R y LaTeX"** by Mario Alfonso Morales Rivera (PDF).
    - **"Metodos Estadisticos con R y R Commander"** by Antonio Jose Saez Castillo (PDF, ZIP, 2010-07-08).
    - **"Optimización Matemática con R: Volumen I"** by Enrique Gabriel Baquela and Andrés Redchuk (PDF, 161 pages).
    - **"Introducción al uso de R y R Commander para el análisis estadístico de datos en ciencias sociales"** by Rosario Collatón Chicana (PDF, 128 pages, 2014-05-11).
    - **"El arte de programar en R"** by Julio Sergio Santana and Efraín Mateos Farfán (PDF, 197 pages, 2014-12-15; online).